

Supplementary Materials

for

Use it or lose it: A model-based assessment of the hypothesis that European Neanderthals relied on wildfires to create their campfires

Authors

Andreu Arinyo-i-Prats^{1,2,3*}, Dennis Sandgathe¹, Felix Riede², Mark Collard^{1*}

*To whom correspondence should be addressed: aarinyoiprats@gmail.com, mcollard@sfu.ca.

<i>Task</i>	<i>Repeat Time [yr]</i>	<i>% Passed</i>	<i>τ [yr]</i>	<i>Source</i>
Salute	1	70	2.8	(67)
Load and Fire	1	55	1.7	(67)
Password	1	42	1.2	(67)
Restore	1	31	0.9	(67)
CPR	1	21	0.6	(67)
Inspect	1	18	0.6	(67)
Wear Gas Mask	1	5	0.3	(67)
Assembly	0.8	1	0.2	(67)
Evaluate Casualty	3	73	9.5	(68)
Prevent Shock	3	65	7.0	(68)
Apply Field Dressing	3	62	6.3	(68)
Wear M-17 Mask	3	37	3.0	(68)
Maintain M-17 Mask	3	34	2.8	(68)
Maintain M16A2 Rifle	3	35	2.9	(68)
Perform Function Check M16	3	24	2.1	(68)
Correct Malfunction M16	3	23	2.0	(68)
Identify Terrain Features	3	28	2.4	(68)
Determine Ground Location	3	27	2.3	(68)
Determine Grid Coordinates	3	22	2.0	(68)
React to Chemical/Biohazard	3	19	1.8	(68)
Decontaminate Skin/Equipment	3	17	1.7	(68)
Ready Reserve	0.3, 0.66, 1	75, 55, 50	1.3	(68)
CPR	0.25, 0.5, 1, 2, 3	40, 18, 14, 8, 0	0.5	(39)
Experiment with five training sessions	0.04, 0.08, 0.5, 1, 2	95, 90, 30, 30, 10	0.8	(69)
Experiment with 30 training sessions	0.04, 0.08, 0.5, 1, 2	0.5, 1, 6, 12, 24	1.8	(69)

Table S1. Estimates of Forgetting Time for different procedural-motor skills. The data in the table were obtained from studies dealing with the decay of procedural-motor skills. The general approach used in the studies was to recruit a group of individuals and train them to perform a task to the point of being able to pass a competency test. After several months the same individuals were recalled and tested on how well they could still perform the task. This second test had a threshold that determined if an individual remained competent. *Task* = a short name for the task. *Repeat Time* = the time in years between the end of training and the second test. *% Passed* = the percentage of the initial participants who passed the second test. *τ* = the estimated rate of decay of the skill. *τ* was computed from *Repeat Time* and *% Passed*, assuming that the decay was exponential and that 100% of individuals passed the initial test. *Source* = the publication from which the data were taken. The data presented in the last four rows derive from studies in which competence was reassessed multiple times, not just once. The two rows that pertain to Ammons et al.'s (69) study require further explanation. These authors taught a procedural-motor experiment to two groups. One group received five training sessions; the other received 30. To demonstrate that it is reasonable to assume exponential decay for procedural-motor skills, the data from the bottom four rows are plotted in S1 Fig, together with error bars and exponential decay curves based on the relevant estimated forgetting times (*τ*).

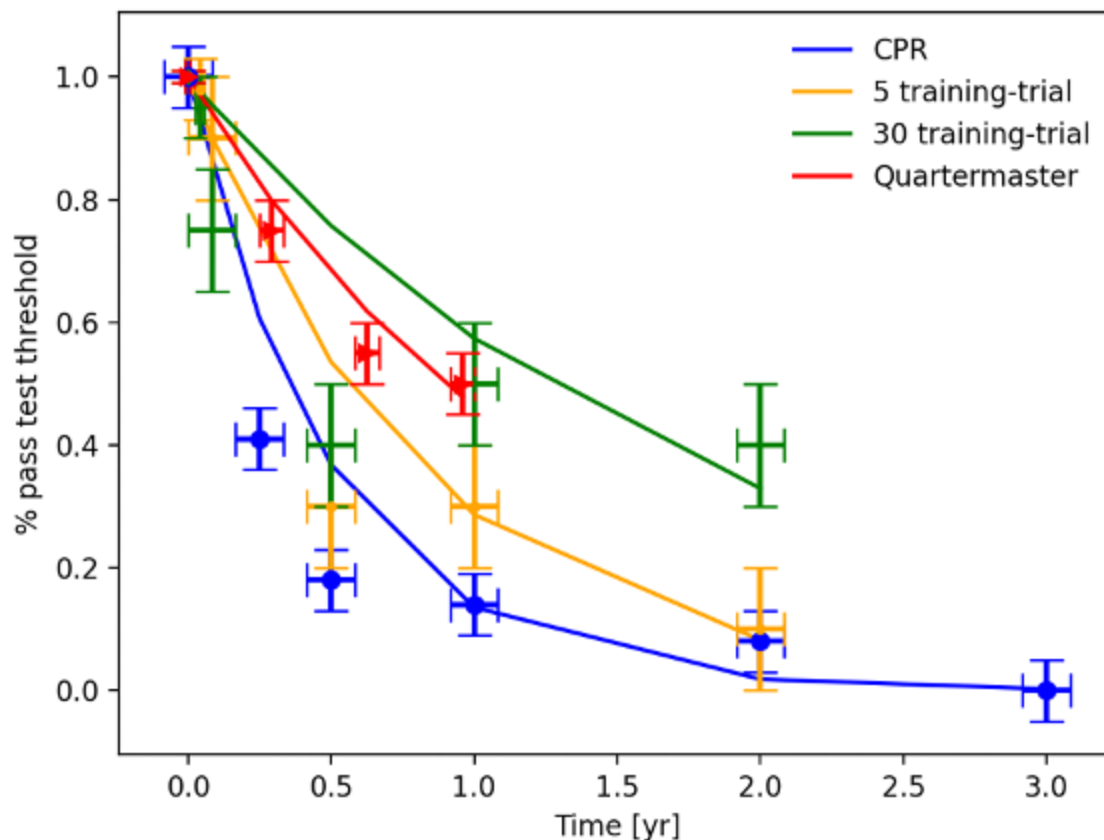


Fig. S1. Plot justifying the assumption that the decay of procedural-motor skills follows an exponential curve. In the plot, we have fitted exponential decay curves to data provided in the four cases we found in the literature that track the decay of procedural-motor skills longer than six months. The values for time and ‘% pass test threshold’ are the same as those found in the last four rows of Table S1. As in Table S1, we assumed that the percentage of skilled individuals was 100% at time 0. The errors shown on the time axis are uncertainties of 1 month. Their equivalents on the ‘% pass test threshold’ axis were visually estimated from published graphs apart from those for CPR. Errors were neither listed in the main text nor included in a plot in the relevant CPR study (39). Accordingly, we added a 5% error to all of the values to account for the inevitable uncertainties.

SM Bibliography

39. A. I. Glendon, S. P. McKenna, K. Hunt, S. S. Blaylock, Variables affecting cardiopulmonary resuscitation skill decay. *J. Occup. Psychol.* **61**, 243–255 (1988).
67. M. A. Sabol, R. A. Wisher, Retention and reacquisition of military skills. *Mil. Oper. Res.* **6**, 59–80 (2001).
68. R. Wisher, M. Sabol, R. Maisano, B. Knott, C. Curnow, J. Ellis, “Retraining medics for mobilization” in *Annual Meeting of the American Educational Research Association*. New York (1996).
69. R. B. Ammons, R. G. Farr, E. Bloch, E. Neumann, M. Dey, R. Marion, C. H. Ammons, Long-term retention of perceptual-motor skills. *J. Exp. Psychol.* **55**, 318–328 (1958).